

1. Which of the following is NOT a criterion used by process scheduling algorithms?

1 point

- ☐ Maximizing CPU utilization
- ☐ Ensuring fairness among processes
- ☒ Reducing main memory size
- ☐ Minimizing turnaround time

2. What is the main role of the medium-term scheduler?

1 point

- ☒ To temporarily remove processes from main memory and place them on secondary memory, or vice versa
- ☐ To allocate CPU to processes in the ready queue
- ☐ To decide which processes should be brought into the ready queue from the job queue
- ☐ To handle the scheduling of processes that are blocked and waiting for I/O resources

3. Consider a five state model. Which state transitions invoke the CPU scheduler?

1 point

- ☐ From ready to blocked, from running to new, from waiting to running, and process termination
- ☐ From waiting to running, from ready to running, from running to blocked, and process suspension
- ☒ From running to waiting, from running to ready, from waiting to ready, and running to termination
- ☐ From new to ready, from ready to running, from running to blocked, and process creation

4. What is the main difference between non-preemptive and preemptive scheduling?

1 point

- ☐ Non-preemptive scheduling allows a running process to be interrupted, while preemptive scheduling does not.
- ☐ Neither allows a running process to be interrupted.
- ☐ Both allow a running process to be interrupted.
- ☒ Non-preemptive scheduling does not allow a running process to be interrupted, while preemptive scheduling does.

5. Consider 5 processes whose Arrival time and CPU Burst time is as follows (Process no: Arrival time: Burst time):

1 point

P1: 0, 3

P2: 2, 3

P3: 3, 2

P4: 5, 5

P5: 6, 9

Assume that the processes are scheduled using the FCFS scheduling algorithm. The average waiting and turnaround time is _____

- ☒ 2.8 and 7.2
- ☐ 1.9 and 6.9
- ☐ 2.5 and 7.0
- ☐ 2.6 and 7.1